	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE **Regular & Supplementary Winter Examination-2023** **Course: B. Tech.** **Branch :Computer and Allied** **Semester :VII** **Subject Code & Name: Deep Learning (BTCOE705B)** **Max Marks: 60** **Date:11/01/2024** **Duration: 3 Hr.**		
	Instructions to the Students: 1. *All the questions are compulsory.* 2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.* 3. *Use of non-programmable scientific calculators is allowed.* 4. *Assume suitable data wherever necessary and mention it clearly.*		
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	**Explain the following steps involved in machine learning.** **(i) Preprocessing (ii)Segmentation (iii)Feature Extraction.**	**Understand**	**6**
B)	**How data is classified using Bayes classifier?**	**Analyze**	**6**
C)	**Distinguish between supervised and unsupervised machine learning.**	**Analyze**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	**What is neuron? Explain structure of biological neuron.**	**Remember**	**6**
B)	**Explain how XOR function is implemented using NAND, OR and AND functions in neural networks.**	**Apply**	**6**
C)	**Explain architecture of feed forward neural network with necessary convention.**	**Understand**	**6**
Q. 3	**Solve Any Two of the following.**		**12**
A)	**What is back propagation learning?**	**Remember**	**6**
B)	**Explain architecture of multilayer perceptron.**	**Remember**	**6**
C)	**How are weights updated at output layer in multilayer neural network?**	**Understand**	**6**
Q.4	**Solve Any Two of the following.**		**12**
A)	**What is Autoencoder? Explain undercomplete and sparse autoencoders in detail.**	**Remember**	**6**
B)	**Explain function of different layers of Convolutional neural networks.**	**Understand**	**6**
C)	**What are the advantages of Convolutional neural networks over multilayer perceptron.**	**Analyze**	**6**

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Q. 5	Solve Any Two of the following.		12
A)	State some application of Deep learning.	Understand	6
B)	What is the need of normalization. Explain batch normalization.	Remember	6
C)	Explain the working of recurrent neural network.	Remember	6
	*** End ***		

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